KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · MATH

Malik's Mural Project

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Before the First Tile

Malik's art class is about to start a giant mural project. Before you meet him, warm up on the exact math skills he'll need today.

1 Match each word to its meaning. Draw a line, or write the matching letter next to each word.

___ **standard algorithm**	**A.** A smart, quick guess made by rounding first.
___ **partial product**	**B.** The stacked, step-by-step way of multiplying larger numbers on paper.
___ **regroup**	**C.** A number being multiplied.
___ **product**	**D.** Carrying extra value into the next place value column.
___ **factor**	**E.** One piece of a multiplication problem, found before adding all the pieces together.
___ **estimate**	**F.** The answer to a multiplication problem.

2 Preview: multiply 14 × 3 using partial products. Break 14 into 10 + 4, multiply each part by 3, then add. Show every step.

3 Have you ever helped with a big group project, like a mural, poster, or art display? Write 1–2 sentences about it.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · MATH

Malik's Mural Project

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Counting Every Tile

Great work on today's lesson! Now show off what you learned about the standard algorithm, partial products, and regrouping.

1. The gym mural was such a hit that Malik's class is designing a SECOND mural for the cafeteria wall: 32 tiles across and 26 tiles down. Use the standard algorithm to find how many tiles it needs. Show every step, including both partial products.

2. Each jar of paint covers 148 tiles. Malik has 6 jars. How many tiles can all 6 jars cover? Use the standard algorithm and show where you regroup. Is that enough paint for the cafeteria mural?

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Malik's Mural Project

Name: _____ Date: _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON · CONTINUED

Counting Every Tile

3 Gold border tiles come in packs of 28, and the class needs 19 packs. First, ESTIMATE the total number of gold tiles by rounding each factor to the nearest ten. Then use the standard algorithm to find the exact product. Was your estimate higher or lower than the exact answer? Why?

4 Malik's classmate multiplied 24 × 18 and got 216. Malik checked his work: the first partial product (24 × 8 = 192) was right, but the second one was written as 24 instead of 240. Explain in 1–2 sentences what mistake his classmate made, and find the correct product.

★ **Challenge:** Design your own tile mural! Choose a 2-digit number of tiles across and a 2-digit number of tiles down. Use the standard algorithm to find the total number of tiles, then estimate to check that your answer makes sense.

Looking Ahead: Next week we divide BIG numbers using 2-digit divisors — Deja needs help splitting her class's food drive cans evenly among 24 pantries! Bring this sheet back to your next lesson.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · READING

Amara's First Solo

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Before Amara Steps Up

Amara has played violin in her school orchestra for two years, always surrounded by other musicians. Now her teacher wants her to play a short solo, alone, in front of the whole school. Before you read her story, warm up your reading brain.

1 Match each word to its meaning. Draw a line, or write the matching letter next to each word.

___ **character change**	**A.** Specific facts or details that support an answer.
___ **turning point**	**B.** How a character's feelings, beliefs, or behavior are different at the end of a story than at the start.
___ **confidence**	**C.** Figuring something out from clues in the text.
___ **gradual**	**D.** A feeling of trust in your own ability to do something.
___ **inference**	**E.** The key moment in a story where a character's choice or feeling starts to shift.
___ **evidence**	**F.** Happening slowly, over time, in small steps — not all at once.

2 Read this sentence: "Amara's hands trembled as she picked up her violin." What can you INFER about how she feels? Write your answer in a full sentence.

3 Has your confidence at something ever grown gradually, over time, instead of all at once? Write 1–2 sentences about it.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · READING

Amara's First Solo

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

After the Final Note

Great work on today's lesson! Now show off what you learned about character change, turning points, and inferring feelings from clues.

1 In 3–4 sentences, retell how Amara CHANGED from the start of the story to the end.

2 What was the TURNING POINT of the story — the moment her feelings started to shift? Explain in 1–2 sentences.

3 Find ONE detail or quote from the story that shows Amara's confidence at the END. Copy it exactly.

★ **Challenge:** Write your own 4–5 sentence story about a character whose confidence grows gradually. Include a clear turning point.

Looking Ahead: Next week we compare TWO different viewpoints on a class vote — get ready to see both sides! Bring this sheet back to your next lesson.

KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · SCIENCE

Weathering, Erosion and Landforms

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. How do you think rocks get broken into sand? Write your best guess.

2. Draw a river flowing from a mountain to the ocean. Label where you think the water moves fastest.

3. Predict: will more soil wash down a steep hill or a gentle hill in a rainstorm? Why?

4. Circle what you think can change solid rock: water, wind, ice, plant roots, sunlight. Explain one choice.

5. What is one thing you wonder about canyons, beaches or sinkholes?

KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · SCIENCE

Weathering, Erosion and Landforms

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use evidence from today's lesson to explain your thinking.

1. Explain the difference between weathering, erosion and deposition. Give one example of each.

2. Use the stream table data: what two things changed how much sand washed away? Give numbers as evidence.

3. Draw and label how a Florida sinkhole forms. Show the rock, the water and the collapse.

4. Compare a canyon and a delta. How can the same river help make both?

★ **Challenge:** Design a fair test to find out whether sand or soil with gravel erodes faster. Name the one variable you will change, two you will keep the same, and what you will measure.

Looking Ahead: Next week we follow water on its trip through the water cycle and learn to read weather data, featuring meteorologist June Bacon-Bercey. Question to bring: Where does the water in a puddle go when it disappears on a sunny day?

KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · SOCIAL STUDIES

The Colonies and Who Built Them

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. What have you heard about Jamestown, Plymouth or Pocahontas? Write two things.

2. Who was already living on the land where the English built their colonies? Write what you know or guess.

3. Prediction: Why might an Indigenous nation choose to trade with colonists? Why might it choose to resist them?

4. Guess: Why would a colony need many workers? What kinds of work would they do?

5. Write one question you have about the first colonies.

KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · SOCIAL STUDIES

The Colonies and Who Built Them

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

What Did I Learn?

Use evidence from today's lesson. Write in complete sentences.

1. Using John Rolfe's letter as evidence, explain how the first Africans came to Virginia in 1619 and what the letter shows about Rolfe's point of view.

2. Choose Matoaka (Pocahontas), Massasoit or Anthony Johnson. Tell that person's story honestly in two or three sentences.

3. Compare William Penn's treaties with the 1737 Walking Purchase. How were they different?

4. Who benefited from Virginia's tobacco, and who paid the cost?

★ **Challenge:** Many old books call Plymouth's 1621 harvest feast "the first Thanksgiving" and show only friendly neighbors. Write a more complete caption for that picture, using two facts from today's lesson.

Looking Ahead: Next week we study how slavery grew in the colonies and how enslaved Africans resisted and kept their cultures alive. Bring this question: how could a colony's laws make slavery last from one generation to the next?

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6A · MATH

Coach's Equipment Count

Name: **Date:**

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Warm Up Your Math Toolbox

Try your best on your own. Show your thinking!

1. Use the area model: 23 × 4 = (20 × 4) + (3 × 4) = _____

SHOW YOUR WORK

2. Estimate 418 + 267 by rounding each number to the nearest hundred first.

3. 36 ÷ 4 = _____. Then write the whole fact family for 4, 9 and 36.

4. What does it mean to *regroup*? Explain in one complete sentence.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6A · MATH

Coach's Equipment Count

Name: **Date:**

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What You Learned

You finished the lesson. Now show every step on paper!

1. Use the standard algorithm: 34 × 7 = _____

SHOW YOUR WORK

2. 26 × 13 = _____. Use the area model or the standard algorithm.

SHOW YOUR WORK

3. Coach Priya has 48 balls to split equally into 6 bins. How many balls go in each bin? Write the fact family that proves it.

4. Coach Priya has 58 cones for 7 drills. ESTIMATE how many cones each drill gets, then explain how you estimated.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6A · MATH

Coach’s Equipment Count

Name: **Date:**

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON · CONTINUED

Show What You Learned

Keep going. Write big and neat!

★ **Challenge:** Write a word problem where division is the “undo button” for a multiplication problem. Solve both.

Looking Ahead: In Week 7 you will learn long division with 2-digit divisors at Deja’s food drive. Bring this sheet to your next lesson.